

Model uncertainty and validation

Exercise

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Do the whiteness test using the data from the KF/EKF exercise where data were simulated with LSim/NLSim.

1. Make one or more whiteness test for combinations of LSim/NLSim and KF/EKF.
2. Make the normal test using normplot as well.
3. What happens if the process and or measurement noise is uniform instead of normal? The LSim.m can simulate this.
4. Explain what happens for increasing number of samples $N = 10, 100, 1000, 10000$.
5. Try to estimate one (or more) parameter e.g. a in the linear system.